

Predictive_PS4(Q1 except d)_735

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PREDICTIVE ANALYTICS

Problem Set 4

1. Problem to demonstrate the fitting of a binary response variable using logistic regression

- Consider the Weekly data available in the ISLR package of R. Use the data from 1990 to 2008 as train data and the data on 2009-2010 as test data.

```
library(ISLR)
```

```
## Warning: package 'ISLR' was built under R version 4.5.2
```

```
data(Weekly)
```

```
str(Weekly)
```

```
## 'data.frame': 1089 obs. of 9 variables:
## $ Year : num 1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 1990 ...
## $ Lag1 : num 0.816 -0.27 -2.576 3.514 0.712 ...
## $ Lag2 : num 1.572 0.816 -0.27 -2.576 3.514 ...
## $ Lag3 : num -3.936 1.572 0.816 -0.27 -2.576 ...
## $ Lag4 : num -0.229 -3.936 1.572 0.816 -0.27 ...
## $ Lag5 : num -3.484 -0.229 -3.936 1.572 0.816 ...
## $ Volume : num 0.155 0.149 0.16 0.162 0.154 ...
## $ Today : num -0.27 -2.576 3.514 0.712 1.178 ...
## $ Direction: Factor w/ 2 levels "Down","Up": 1 1 2 2 2 1 2 2 2 1 ...
```

```
train=Weekly[Weekly$Year<=2008,]
```

```
test =Weekly[Weekly$Year>2008,]
```

```
nrow(train)
```

```
## [1] 985
```

```
nrow(test)
```

```
## [1] 104
```

- Using the training data, fit a logistic regression with Direction as the response and the five lag variables plus Volume as predictors. Are there any significant predictors? If not, which of the tests is reporting the least p value. Interpret the coefficient corresponding to that test.

```

log_reg=glm(Direction ~ Lag1+Lag2+Lag3+Lag4+Lag5+Volume,
             data=train,
             family=binomial)

summary(log_reg)

##
## Call:
## glm(formula = Direction ~ Lag1 + Lag2 + Lag3 + Lag4 + Lag5 +
##      Volume, family = binomial, data = train)
##
## Coefficients:
##              Estimate Std. Error z value Pr(>|z|)
## (Intercept)  0.33258    0.09421   3.530 0.000415 ***
## Lag1        -0.06231    0.02935  -2.123 0.033762 *
## Lag2         0.04468    0.02982   1.499 0.134002
## Lag3        -0.01546    0.02948  -0.524 0.599933
## Lag4        -0.03111    0.02924  -1.064 0.287241
## Lag5        -0.03775    0.02924  -1.291 0.196774
## Volume      -0.08972    0.05410  -1.658 0.097240 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## (Dispersion parameter for binomial family taken to be 1)
##
##      Null deviance: 1354.7  on 984  degrees of freedom
## Residual deviance: 1342.3  on 978  degrees of freedom
## AIC: 1356.3
##
## Number of Fisher Scoring iterations: 4

coef(log_reg)["Lag2"]

##      Lag2
## 0.04467963

```

c. Estimate the odds of market going “up” at median values of the predictors.

```

median=apply(train[,c("Lag1", "Lag2", "Lag3", "Lag4", "Lag5", "Volume")],
             2, median)

median_df=as.data.frame(t(median))

prob=predict(log_reg, newdata=median_df, type="response")

odds=prob/(1-prob)

prob

##      1
## 0.5589808

```

```
odds
```

```
##          1  
## 1.267475
```

- e. Using the fitted model and the predictor values from the test data, find an optimal cut point that maximizes $TPR(1-FPR)$. Report the confusion matrix corresponding to the optimal cut-point value. From the confusion matrix compute the test error rate.

```
test_prob=predict(log_reg,newdata=test,type="response")
```

```
library(pROC)
```

```
## Warning: package 'pROC' was built under R version 4.5.3
```

```
## Type 'citation("pROC")' for a citation.
```

```
##
```

```
## Attaching package: 'pROC'
```

```
## The following objects are masked from 'package:stats':
```

```
##
```

```
##      cov, smooth, var
```

```
roc_obj=roc(test$Direction,test_prob)
```

```
## Setting levels: control = Down, case = Up
```

```
## Setting direction: controls < cases
```

```
tpr=roc_obj$sensitivities
```

```
fpr=1-roc_obj$specificities
```

```
threshold=roc_obj$threshold
```

```
metric=tpr*(1-fpr)
```

```
opt_index=which.max(metric)
```

```
opt_cutoff=threshold[opt_index]
```

```
opt_cutoff
```

```
## [1] 0.4568085
```

```
test_pred=ifelse(test_prob>opt_cutoff,"Up","Down")
```

```
table(Predicted=test_pred,Actual=test$Direction)
```

```
##          Actual
```

```
## Predicted Down Up
```

```
##      Down   22 27
```

```
##      Up    21 34
```

```
mean(test_pred!=test$Direction)
```

```
## [1] 0.4615385
```

- f. Is there any change in the test error rate if we have worked with only the two predictors lag 1 and lag 2?

```
red_fit=glm(Direction ~ Lag1+Lag2,  
             data=train,  
             family=binomial)
```

```
summary(red_fit)
```

```
##
```

```
## Call:
```

```
## glm(formula = Direction ~ Lag1 + Lag2, family = binomial, data = train)
```

```
##
```

```
## Coefficients:
```

```
##              Estimate Std. Error z value Pr(>|z|)  
## (Intercept)  0.21109    0.06456   3.269  0.00108 **  
## Lag1        -0.05421    0.02886  -1.878  0.06034 .  
## Lag2         0.05384    0.02905   1.854  0.06379 .
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
```

```
## (Dispersion parameter for binomial family taken to be 1)
```

```
##
```

```
## Null deviance: 1354.7 on 984 degrees of freedom
```

```
## Residual deviance: 1347.0 on 982 degrees of freedom
```

```
## AIC: 1353
```

```
##
```

```
## Number of Fisher Scoring iterations: 4
```

```
red_test_prob=predict(red_fit,newdata=test,type="response")
```

```
red_roc_obj=roc(test$Direction,red_test_prob)
```

```
## Setting levels: control = Down, case = Up
```

```
## Setting direction: controls < cases
```

```
red_tpr=red_roc_obj$sensitivities
```

```
red_fpr=1-red_roc_obj$specificities
```

```
red_threshold=red_roc_obj$threshold
```

```
red_metric=red_tpr*(1-red_fpr)
```

```
red_opt_cutoff=red_threshold[which.max(red_metric)]
```

```
red_test_pred=ifelse(red_test_prob>red_opt_cutoff,"Up","Down")
```

```
table(Predicted=red_test_pred,Actual=test$Direction)
```

```
##           Actual
## Predicted Down Up
##      Down   19 13
##      Up     24 48

mean(red_test_pred != test$Direction)

## [1] 0.3557692
```